Table 1: arch sweep: Enzyme/2 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
GRU	–	–	0.046	–
LSTM	–	–	0.079	–
sLSTM	**0.533**	**0.114**	0.064	**0.012**
Transformer	–	–	**0.042**	–
Mamba	–	–	0.160	–
First–last observation				
GRU	–	–	**0.046**	–
LSTM	–	–	–	–
sLSTM	–	–	–	–
Transformer	–	–	–	–
Mamba	–	–	–	–

Table 2: arch sweep: Glyc/12 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
GRU	0.610	0.266	0.148	**0.072**
LSTM	0.371	**0.191**	0.395	0.083
sLSTM	0.338	0.312	0.143	0.077
Transformer	**0.222**	0.221	**0.119**	0.073
Mamba	0.249	0.257	0.162	0.125
First–last observation				
GRU	**0.309**	0.376	0.508	0.478
LSTM	0.384	**0.319**	0.508	**0.393**
sLSTM	0.347	0.793	0.551	0.465
Transformer	0.396	0.359	0.416	0.449
Mamba	0.333	0.455	**0.341**	0.772

Table 3: arch sweep: Glyc/8 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
GRU	0.146	0.112	0.129	**0.113**
LSTM	0.269	0.132	**0.122**	8.0×10^{6}
sLSTM	0.184	0.111	–	–
Transformer	**0.117**	0.114	0.129	0.123
Mamba	0.200	**0.108**	0.130	0.168
First–last observation				
GRU	0.250	0.485	0.742	0.887
LSTM	0.206	**0.300**	0.736	0.939
sLSTM	0.154	0.704	–	–
Transformer	0.201	0.537	0.631	**0.821**
Mamba	**0.131**	0.387	**0.564**	0.880

Table 4: arch sweep: MOF/4 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
GRU	0.181	0.155	0.119	0.478
LSTM	0.163	0.151	0.125	0.487
sLSTM	0.188	0.155	0.124	**0.438**
Transformer	**0.147**	**0.126**	**0.100**	0.440
Mamba	0.222	0.152	0.117	1.146
First–last observation				
GRU	1.050	0.540	0.766	12.024
LSTM	0.935	0.681	0.788	19.292
sLSTM	0.848	**0.414**	**0.321**	**7.404**
Transformer	0.831	0.585	0.774	10.535
Mamba	**0.798**	0.693	0.569	7.0×10^{12}

Table 5: arch sweep: MOF/6 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
GRU	–	**0.067**	0.057	**0.039**
LSTM	–	0.080	0.058	0.044
sLSTM	**0.122**	0.085	**0.056**	0.067
Transformer	–	–	0.060	0.049
Mamba	–	–	0.065	0.163
First–last observation				
GRU	–	0.130	0.121	**0.154**
LSTM	–	–	**0.116**	0.162
sLSTM	**0.351**	**0.130**	0.176	0.173
Transformer	–	–	0.129	0.199
Mamba	–	–	0.178	0.339

Table 6: data ablation: Enzyme/2 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
CMVF	0.526	0.148	0.046	0.015
CMVF-L1	0.526	0.148	**0.044**	0.016
CMVF-L2	–	–	–	0.016
CMVF-unbounded	0.606	**0.080**	0.055	**0.012**
NODE-GRU	0.776	0.103	0.083	0.020
NODE-MLP	0.729	0.116	0.181	0.319
NODE-correction	**0.424**	0.083	0.177	0.305
Global (θ)	0.466	0.348	0.241	0.413
Initial-condition (θ)	0.500	0.180	0.266	0.422
First–last observation				
CMVF	0.526	0.148	0.046	–
CMVF-L1	0.526	0.148	**0.044**	–
CMVF-L2	–	–	–	–
CMVF-unbounded	0.606	**0.080**	0.055	–
NODE-GRU	0.776	0.103	0.083	–
NODE-MLP	0.729	0.116	0.181	–
NODE-correction	**0.424**	0.083	0.177	–
Global (θ)	0.466	0.348	0.241	**0.413**
Initial-condition (θ)	0.500	0.180	0.266	0.422

Table 7: data ablation: Enzyme/4 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
CMVF	0.031	0.074	**0.005**	0.004
CMVF-L1	0.038	0.041	0.009	0.007
CMVF-unbounded	0.046	0.032	0.015	**0.004**
NODE-GRU	0.733	0.139	0.031	0.011
NODE-MLP	0.509	0.184	0.042	0.006
NODE-correction	0.039	0.031	0.035	1.1×10^6
Global (θ)	0.031	0.030	0.036	0.042
Initial-condition (θ)	**0.030**	**0.028**	0.036	1.1×10^6
First–last observation				
CMVF	**0.030**	0.060	**0.005**	**0.004**
CMVF-L1	0.038	0.060	0.009	0.007
CMVF-unbounded	0.037	**0.024**	0.006	1.1×10^6
NODE-GRU	0.520	0.478	0.305	0.257
NODE-MLP	0.279	0.193	0.265	0.291
NODE-correction	0.067	0.050	0.103	0.086
Global (θ)	0.031	0.031	0.037	0.042
Initial-condition (θ)	0.031	0.028	0.036	1.1×10^6

Table 8: data ablation: Enzyme/6 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
CMVF	0.013	0.007	**0.003**	**0.002**
CMVF-L1	**0.007**	0.010	0.008	0.005
CMVF-unbounded	0.011	0.005	0.004	0.002
NODE-GRU	4.3×10^6	1.9×10^6	6.2×10^5	1.6×10^5
NODE-MLP	7.2×10^6	1.0×10^6	4.1×10^5	6.9×10^4
NODE-correction	3.0×10^6	2.4×10^5	1.0×10^5	1.3×10^4
Global (θ)	0.019	0.014	0.014	0.007
Initial-condition (θ)	0.009	**0.002**	0.003	0.004
First–last observation				
CMVF	0.012	0.009	**0.003**	**0.003**
CMVF-L1	0.010	0.010	0.004	0.005
CMVF-unbounded	0.012	0.005	0.005	0.003
NODE-GRU	5.4×10^7	6.6×10^7	1.6×10^7	2.2×10^7
NODE-MLP	1.4×10^7	5.0×10^7	1.9×10^7	2.7×10^7
NODE-correction	7.7×10^6	2.4×10^7	2.1×10^7	1.5×10^7
Global (θ)	0.020	0.021	0.014	0.008
Initial-condition (θ)	**0.009**	**0.002**	0.003	0.004

Table 9: data ablation: Glyc/12 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
CMVF	0.610	0.266	**0.148**	**0.072**
CMVF-L1	0.259	0.308	0.161	0.077
CMVF-unbounded	**0.215**	0.266	0.172	0.088
NODE-GRU	0.282	0.314	0.196	0.178
NODE-MLP	0.252	0.312	0.221	0.143
NODE-correction	0.266	0.266	0.166	0.174
Global (θ)	0.334	0.609	0.413	0.184
Initial-condition (θ)	0.253	**0.211**	0.175	0.182
First–last observation				
CMVF	0.309	0.376	0.508	0.478
CMVF-L1	0.307	0.323	0.449	9.4×10^6
CMVF-unbounded	**0.300**	0.644	0.593	**0.265**
NODE-GRU	1.627	2.464	1.339	7.157
NODE-MLP	2.120	2.667	3.563	86.196
NODE-correction	2.021	2.678	1.361	5.618
Global (θ)	0.333	0.553	**0.356**	0.387
Initial-condition (θ)	0.353	**0.272**	0.525	0.674

Table 10: data ablation: Glyc/22 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
CMVF	0.375	0.069	0.021	1.2×10^{13}
CMVF-L1	0.812	0.069	0.027	0.006
CMVF-unbounded	**0.095**	**0.041**	**0.016**	**0.004**
NODE-GRU	562.012	663.087	1.7×10^4	8.0×10^3
NODE-MLP	2.337	3.1×10^5	1.2×10^4	7.1×10^3
NODE-correction	0.285	0.231	5.1×10^3	–
Global (θ)	0.397	0.343	0.299	0.042
Initial-condition (θ)	0.310	0.112	0.030	0.015
First–last observation				
CMVF	0.638	0.273	**0.185**	0.215
CMVF-L1	0.634	0.387	0.221	0.254
CMVF-unbounded	0.564	0.423	0.296	**0.108**
NODE-GRU	6.7×10^6	1.4×10^7	1.2×10^7	9.3×10^6
NODE-MLP	7.0×10^6	3.0×10^6	1.9×10^6	1.0×10^8
NODE-correction	0.436	1.768	5.2×10^6	–
Global (θ)	**0.434**	0.388	0.413	–
Initial-condition (θ)	0.809	**0.270**	0.294	–

Table 11: data ablation: Glyc/4 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
CMVF	**0.234**	0.226	**0.111**	**0.077**
CMVF-L1	0.236	0.253	0.140	0.079
CMVF-unbounded	0.261	0.545	0.258	0.088
NODE-GRU	0.239	0.165	0.145	0.108
NODE-MLP	0.268	0.141	0.178	0.146
NODE-correction	0.295	**0.100**	0.125	0.137
Global (θ)	0.480	0.876	0.556	0.284
Initial-condition (θ)	0.267	0.225	0.215	0.294
First–last observation				
CMVF	0.493	1.247	1.891	4.511
CMVF-L1	0.347	0.859	1.094	1.697
CMVF-unbounded	2.589	2.058	1.669	28.351
NODE-GRU	**0.329**	**0.438**	**0.388**	**0.273**
NODE-MLP	0.532	0.572	0.523	7.494
NODE-correction	0.378	1.114	1.102	0.348
Global (θ)	0.565	1.062	0.909	0.892
Initial-condition (θ)	0.523	0.775	1.022	0.934

Table 12: data ablation: Glyc/8 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
CMVF	**0.146**	0.112	0.129	0.113
CMVF-L1	0.181	0.146	0.149	0.101
CMVF-unbounded	0.192	**0.078**	**0.119**	**0.082**
NODE-GRU	0.169	0.135	0.135	0.141
NODE-MLP	0.271	0.112	0.122	0.133
NODE-correction	0.247	0.169	0.136	0.140
Global (θ)	0.569	1.015	0.960	0.339
Initial-condition (θ)	0.182	0.442	0.254	0.273
First–last observation				
CMVF	0.250	0.485	0.742	**0.887**
CMVF-L1	0.279	0.772	0.752	1.275
CMVF-unbounded	0.251	0.593	**0.699**	0.953
NODE-GRU	0.377	0.419	0.914	0.893
NODE-MLP	0.311	**0.378**	2.009	5.701
NODE-correction	0.902	2.396	2.288	6.509
Global (θ)	0.564	1.013	0.860	1.739
Initial-condition (θ)	**0.161**	0.461	3.689	4.234

Table 13: data ablation: MOF/12 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
CMVF	**0.086**	**0.014**	7.430	2.144
CMVF-L1	0.167	0.027	38.581	**2.113**
CMVF-unbounded	0.122	0.021	11.565	2.866
NODE-GRU	0.450	0.339	9.7×10^5	3.2×10^4
NODE-MLP	0.543	0.272	1.0×10^7	3.4×10^4
NODE-correction	0.200	0.111	5.3×10^4	5.9×10^3
Global (θ)	0.193	0.160	1.0×10^3	122.809
Initial-condition (θ)	0.094	0.025	**5.848**	2.269
First–last observation				
CMVF	**0.119**	0.112	5.486	3.049
CMVF-L1	0.119	0.113	23.798	1.174
CMVF-unbounded	0.173	**0.080**	21.744	8.517
NODE-GRU	3.777	3.715	1.6×10^7	7.8×10^6
NODE-MLP	1.358	2.121	1.9×10^7	4.3×10^6
NODE-correction	0.185	0.314	2.6×10^5	7.7×10^4
Global (θ)	0.169	0.167	930.252	83.288
Initial-condition (θ)	0.142	0.141	**2.412**	**1.043**

Table 14: data ablation: MOF/4 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
CMVF	0.181	0.155	**0.119**	0.478
CMVF-L1	0.210	0.147	0.121	0.672
CMVF-L2	–	–	–	0.490
CMVF-unbounded	0.534	0.133	0.138	**0.428**
NODE-GRU	0.407	0.159	2.0×10^4	2.1×10^4
NODE-MLP	1.045	0.132	1.3×10^6	5.0×10^5
NODE-correction	0.401	**0.131**	0.164	–
Global (θ)	0.593	0.501	0.330	4.018
Initial-condition (θ)	**0.178**	0.159	0.135	3.413
First–last observation				
CMVF	1.050	0.540	**0.766**	12.024
CMVF-L1	1.043	0.602	11.351	9.588
CMVF-L2	–	–	–	–
CMVF-unbounded	**0.735**	**0.424**	23.357	**3.835**
NODE-GRU	1.989	3.658	8.8×10^5	1.6×10^7
NODE-MLP	0.871	3.890	2.9×10^7	1.2×10^7
NODE-correction	0.979	0.865	2.9×10^5	1.2×10^5
Global (θ)	0.948	0.668	0.859	16.990
Initial-condition (θ)	0.743	0.571	0.833	15.714

Table 15: data ablation: MOF/6 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
CMVF	0.156	**0.067**	**0.057**	**0.039**
CMVF-L1	0.250	0.075	0.057	0.074
CMVF-unbounded	0.196	0.079	0.537	0.126
NODE-GRU	0.393	0.181	1.9×10^5	1.7×10^4
NODE-MLP	0.408	0.182	1.9×10^5	5.2×10^4
NODE-correction	0.317	0.140	0.076	1.4×10^4
Global (θ)	0.240	0.255	0.109	0.236
Initial-condition (θ)	**0.111**	0.154	0.094	0.246
First–last observation				
CMVF	0.338	**0.130**	**0.121**	**0.154**
CMVF-L1	0.273	0.144	0.130	0.459
CMVF-unbounded	0.264	0.132	8.915	0.223
NODE-GRU	1.540	1.173	1.6×10^7	7.4×10^6
NODE-MLP	5.559	16.193	3.5×10^7	5.6×10^6
NODE-correction	0.328	0.309	1.1×10^4	4.7×10^5
Global (θ)	**0.229**	0.240	0.133	0.353
Initial-condition (θ)	0.265	0.172	0.191	0.213

Table 16: data ablation: MOF/8 — NRMSE (mean).

Model	n=3	n=10	n=100	n=1000
Full observation				
CMVF	0.215	**0.052**	**0.021**	**0.015**
CMVF-L1	0.303	0.114	0.029	0.020
CMVF-unbounded	**0.200**	0.118	0.860	0.074
NODE-GRU	0.370	0.246	7.6×10^4	3.0×10^3
NODE-MLP	0.388	0.216	4.4×10^5	1.1×10^4
NODE-correction	0.346	0.173	4.5×10^4	1.0×10^4
Global (θ)	0.300	0.265	0.160	0.095
Initial-condition (θ)	0.249	0.130	0.094	0.095
First–last observation				
CMVF	0.265	**0.151**	**0.070**	**0.042**
CMVF-L1	0.289	0.156	0.170	0.250
CMVF-unbounded	0.268	0.152	1.496	0.093
NODE-GRU	3.369	3.087	1.2×10^8	2.6×10^7
NODE-MLP	3.183	10.429	4.3×10^7	1.9×10^7
NODE-correction	0.409	0.332	1.7×10^5	1.1×10^6
Global (θ)	0.272	0.259	0.168	0.150
Initial-condition (θ)	**0.240**	0.160	0.216	0.202